Orthogonal Sets, $\vec{\S}$6.2, 6.3

Recall from last class:

- Let $u, v \in \mathbb{R}^n$, then the **dot product** of u and v is given by
$$u \cdot v = u^T v = u_1 v_1 + \cdots + u_n v_n.$$
- **Theorem 6.1:** Let $u, v, w \in \mathbb{R}^n$ and $c \in \mathbb{R}$.
  1. $$u \cdot v = v \cdot u.$$
  2. $$(u + v) \cdot w = u \cdot w + v \cdot w.$$
  3. $$(cu) \cdot v = c(u \cdot v) = u \cdot (cv).$$
  4. $u \cdot u \geq 0$ and $u \cdot u = 0$ if and only if $u = 0$.
- The **length** or **norm** of the vector v is a non-negative scalar, denoted by $\|v\|$, and defined by
$$\|v\| = \sqrt{v \cdot v}.$$
- We define the **distance** between u and v by
$$\text{dist}(u, v) = \|u - v\|.$$
- The vectors u and v are **orthogonal** if $u \cdot v = 0$.
- **Theorem 6.2:** The vectors u and v are orthongal if and only if
$$\|u\|^2 + \|v\|^2 = \|u + v\|^2.$$
- The set
$$W^\perp = \{z \in \mathbb{R}^n : z \cdot w = 0 \text{ for all } w \in W\}$$
is called the **orthogonal complement** of W.
- Note: $x \in W^\perp$ if and only if x is orthogonal to each vector in a set that spans W. Also $W^\perp$ is a subspace of $\mathbb{R}^n$.
- We can use the dot product to find the angles between vectors; namely
$$u \cdot v = \|u\| \|v\| \cos \theta.$$
- A set of vectors $\{u_1, \ldots, u_p\}$ in $\mathbb{R}^n$ is **orthogonal** if $u_k \cdot u_j = 0$ for all $j \neq k$.
- **Theorem 6.3** Let A be an $m \times n$ matrix, then
$$(\text{Row}(A))^\perp = \text{Null}(A) \text{ and } (\text{Col}(A))^\perp = \text{Null}(A^T).$$
- **Theorem 6.4:** If $S = \{u_1, \ldots, u_p\}$ is an orthogonal set of non-zero vectors in $\mathbb{R}^n$, then S is linearly independent and hence a basis for Span S.

Definition 1. *An* **orthogonal basis** *for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.*

Theorem 6.5: Let $\{u_1, \ldots, u_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $y \in W$, then

$$y = c_1 u_1 + \cdots + c_p u_p$$

where

$$c_j = \frac{y \cdot u_j}{u_j \cdot u_j}.$$

1. Prove the Theorem.

2. Let $u_1 = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ and $u_2 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$ and $x = \begin{bmatrix} 9 \\ -7 \end{bmatrix}$. Find x in the $\{u_1, u_2\}$-coordinates.

Definition 2. *Let u be a non-zero vector in $\mathbb{R}^n$ and let v be a vector in $\mathbb{R}^n$, then the* **orthogonal projection** *of v onto u, denoted by $proj_u v$, is given by*

$$proj_u v = \frac{v \cdot u}{u \cdot u} u$$

Let $L = Span\{u\}$, then we also write $proj_L v$ for the orthogonal projection of v onto the line defined by the span of u; namely

$$proj_L v = proj_u v.$$

3. Let $u = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$. Find the orthogonal projection of y onto u.

Note: Theorem 6.5 says that

$$y = \text{proj}_{u_1} y + \text{proj}_{u_2} y + \cdots + \text{proj}_{u_p} y.$$

Definition 3. *A set $\{u_1, \ldots, u_p\}$ is* **orthonormal** *if it is orthogonal and $\|u_k\| = 1$ for all k.*

Theorem 6.6: An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I_n$.

4. Prove the Theorem.

Theorem 6.7: Let U be an $m \times n$ matrix with orthonormal columns. Let $x, y \in \mathbb{R}^n$. Then

(a) $\|Ux\| = \|x\|$.

(b) $(Ux) \cdot (Uy) = x \cdot y$.

(c) $(Ux) \cdot (Uy) = 0$ if and only if $x \cdot y = 0$.

5. Prove the theorem.

Definition 4. *An* **orthogonal matrix** U *is a square matrix so that* $U^{-1} = U^T$.

Note: An $n \times n$ matrix U is orthogonal if and only if the columns of U are orthonormal.

6. Suppose U is an orthogonal matrix.

(a) What can you say about the eigenvalues of U?

(b) What is the determinant of U?

Orthogonal Sets – Answers / Solutions

1. *Proof.* Suppose $y = c_1u_1 + \ldots c_pu_p$. Now take the dot product of both sides with u_k to get

$$y \cdot u_k = (c_1u_1 + \cdots + c_pu_p) \cdot u_k.$$

Using the orthogonality of $u_1, \ldots, u_p$, we can simplify the right hand side to get

$$y \cdot u_k = c_k u_k \cdot u_k.$$

Therefore

$$c_k = \frac{y \cdot u_k}{u_k \cdot u_k}.$$

□

2. In order to find x in the u_1, u_2 coordinates, we need to find scalars c_1, c_2 such that

$$x = c_1u_1 + c_2u_2.$$

Notice that $u_1 \cdot u_2 = 0$, and hence we can use the previous theorem to conclude

$$c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1} = \frac{39}{13} = 3,$$

and

$$c_2 = \frac{x \cdot u_2}{u_2 \cdot u_2} = \frac{26}{52} = \frac{1}{2}.$$

Thus $[x]_{\{u_1,u_1\}} = \begin{bmatrix} 3 \\ \frac{1}{2} \end{bmatrix}$.

3. The orthogonal projection of y onto u is given by

$$\text{proj}_u y = \frac{y \cdot u}{u \cdot u} u = \frac{10}{8} u = \begin{bmatrix} \frac{5}{2} \\ \frac{5}{2} \end{bmatrix}.$$

4. *Proof.* Let $U = [u_1, \ldots, u_n]$, then

$$U^TU = \begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix} \begin{bmatrix} u_1 & \ldots & u_n \end{bmatrix} = \begin{bmatrix} u_1^Tu_1 & u_1^Tu_2 & \ldots & u_1^Tu_n \\ u_2^Tu_1 & u_2^Tu_2 & \ldots & u_2^Tu_n \\ \vdots & & \ddots & \\ u_n^Tu_1 & u_n^Tu_2 & \ldots & u_n^Tu_n \end{bmatrix} = \begin{bmatrix} u_1 \cdot u_1 & u_1 \cdot u_2 & \ldots & u_1 \cdot u_n \\ u_2 \cdot u_1 & u_2 \cdot u_2 & \ldots & u_2 \cdot u_n \\ \vdots & & \ddots & \\ u_n \cdot u_1 & u_n \cdot u_2 & \ldots & u_n \cdot u_n \end{bmatrix}.$$

Thus $U^TU = I_n$ if and only if $u_k \cdot u_j = 0$ for all $k \neq j$ and $u_k \cdot u_k = 1$. Thus $U^TU = I_n$ if and only if the columns of U are orthonormal. □

5. *Proof.* Let U be an $m \times n$ matrix with orthonormal columns. We prove (b) first.

(b). By the previous theorem we know that $U^TU = I_n$. Now using the definition of the dot product and properties of the transpose of a matrix, we get

$$(Ux) \cdot (Uy) = (Ux)^TUy = x^TU^TUy = x^Ty = x \cdot y.$$

(a). By definition of the norm, $\|Ux\|^2 = (Ux) \cdot (Ux)$. From part (a), we get $\|Ux\|^2 = x \cdot x = \|x\|^2$. Taking square roots, we get the result.

(c). From part (b), $x \cdot y = 0$ if and only if $(Ux) \cdot (Uy) = 0$. □

6. (a) Suppose λ is an eigenvalue of an orthogonal matrix U. Since orthogonal matrices have orthonormal columns, we know that $U^T U = I_n$. Thus

$$\|x\| = \|Ux\| = \|\lambda x\| = |\lambda| \, \|x\| \, .$$

Hence we get $|\lambda| = 1$.

(b) Using the multiplicativity of the determinant we get

$$1 = \det I_n = \det U^T U = \det U^T \det U = (\det U)^2.$$

Thus $\det U = 1$ or $\det U = -1$.